

STA 5635 HW 9

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March 12, 2026

Problem 1

x_min	y_min	Expresssion 1	Expression 2	Expression 3
0.682800	0.521800	0.172200	0.172200	0.172200

Table 1: Minimized x and y values for the three expressions, as well as the values of the expressions.

Problem 2

Part a

Viterbi sequence:

[illegible]

Part b

The ratio of alpha values at time 118: 0.4970

The ratio of beta values at time 118: 1.3175

Problem 3

$$\hat{\pi} = \begin{bmatrix} 1.0000 & 0.0000 \end{bmatrix}$$

$$\hat{A} = \begin{bmatrix} 0.8769 & 0.1231 \\ 0.2160 & 0.7840 \end{bmatrix}$$

$$\hat{B} = \begin{bmatrix} 0.2111 & 0.2120 & 0.2033 & 0.1899 & 0.0914 & 0.0923 \\ 0.1030 & 0.0890 & 0.1118 & 0.2091 & 0.1886 & 0.2984 \end{bmatrix}$$